PRECALCULUS · UNIT 3: CONICS

Science Museum Conics

Design a science museum exhibit with a parabolic dish, an elliptical whispering gallery, and a hyperbolic tower.

CP2

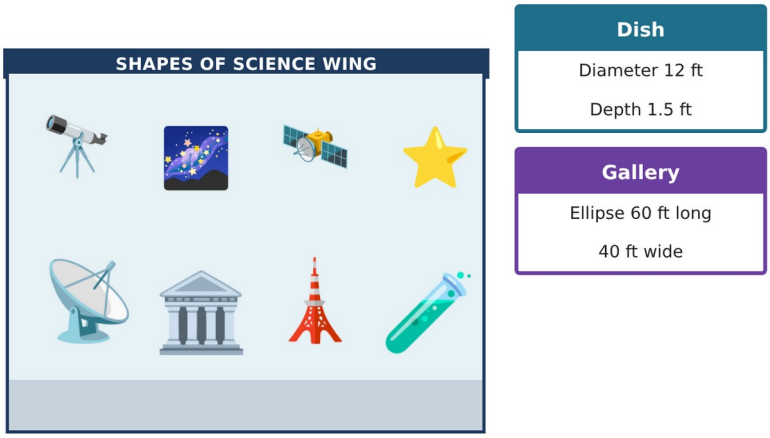
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The city science museum is adding a "Shapes of Science" wing. Each exhibit is built around a conic section: a round planetarium, a parabolic dish, an elliptical whispering room, and a hyperbolic cooling tower model.

You will write the equations, locate the foci, and make the measurements the builders need.



Learning Goals

- I can write and graph the equation of a circle.
- I can find the focus of a parabola.
- I can identify ellipses and hyperbolas from equations.

Project Timeline

When	What you will do
Day 1	Planetarium Circle
Day 2	Parabolic Dish
Day 3	Ellipse & Hyperbola
Final	Exhibit Design Packet + reflection

Part 1: Planetarium Circle

Day 1 · Use the Specification Cards.

Exhibit Specification Cards

Planetarium	Satellite Dish	Flashlight
Circle, center (0, 0)	Diameter 12 ft	Parabolic mirror
radius 25 ft	depth 1.5 ft	$y = \frac{1}{8}x^2$

Whispering Gallery	Cooling Tower	Navigation
Ellipse 60 ft by 40 ft	Hyperbola, narrowest	Two stations
	width 30 ft	100 mi apart

1. Write the equation of the planetarium floor (center (0, 0), radius 25 ft).

Starter: $x^2 + y^2 = ___^2$

2. Is the point (15, 20) on the edge of the floor? Show your check.

3. The projector is moved to (3, −2). Write the equation of a circle of radius 25 centered there.

4. Find the center and radius of $(x + 1)^2 + (y - 4)^2 = 49$.

Part 2: Parabolic Dish

Day 2 · A parabola $x^2 = 4py$ has its focus at $(0, p)$.

5. The flashlight mirror is $y = \frac{1}{8}x^2$, or $x^2 = 8y$. Find p and the focus.

6. Where should the bulb go? Why?

7. The dish passes through (6, 1.5). Find p using $x^2 = 4py$.

8. Write the equation of the dish.

Part 3: Ellipse & Hyperbola

Day 3 · Identify each conic.

9. Identify the conic: $\frac{x^2}{900} + \frac{y^2}{400} = 1$. What are its length and width?

10. Identify the conic: $\frac{x^2}{225} - \frac{y^2}{400} = 1$.

11. Sketch your favorite exhibit and label its key points.

Final Product: Exhibit Design Packet

Your final product must include:

- Circle equations for the planetarium
- Dish and flashlight equations with foci
- Conics identified
- A labeled exhibit sketch
- Exhibit packet

Reflection

R1. Why is the receiver of a satellite dish placed at the focus?

R2. How can you tell a hyperbola from an ellipse in an equation?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Conic equations, foci, and measurements are correct and used to design working exhibits. Goes beyond the minimum with a creative, realistic choice.	Conic equations, foci, and measurements are correct and used to design working exhibits.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Exhibit packet includes equations, labeled sketches, and builder measurements. Polished, organized, and visually engaging.	Exhibit packet includes equations, labeled sketches, and builder measurements. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.